Do Now · retrieval

Lesson 086 · Debugging Unfamiliar Algorithms

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a loop invariant?

Q2

What is a boundary test?

Q3

What return value commonly indicates a failed indexed search?

Q4

In array [2,4,6], set a[1]=a[0]+a[2]. Give the final array.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — Trace before changing code: list variables, loop indexes, condition and expected return.

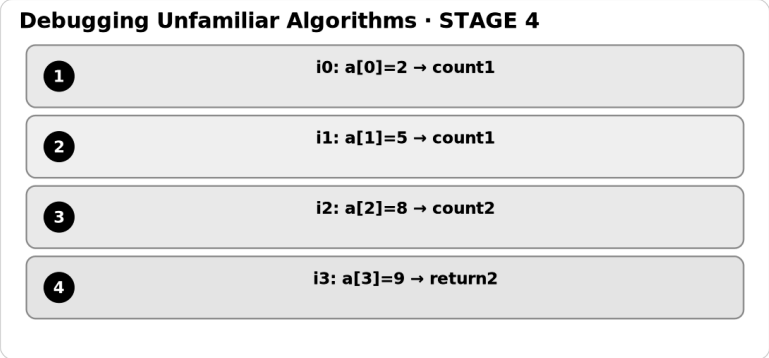
Core idea 2

Explain — An off-by-one fault often comes from <= Length instead of < Length; an empty array must also be considered.

Core idea 3

Evaluate — Correct code by explaining the cause and showing a fresh test, not just copying a working version.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

logic error — Program runs but produces wrong answer.

boundary — Edge of valid range.

trace — Record changing states.

debugging — Finding and correcting faults.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

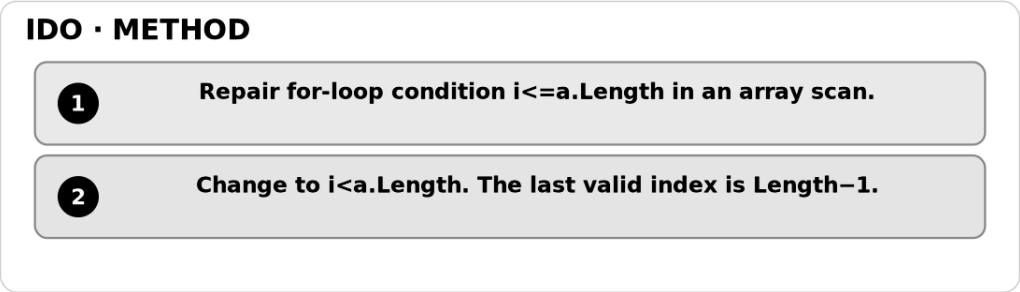
Problem

Repair for-loop condition i<=a.Length in an array scan.

Follow the worked method

1. Model response — For length4, valid indexes0–3. Checking a[4] would throw.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Model response
Initialising to1 incorrectly counts an item before any check.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace CountEven([2,5,8,9]) with count after each item. [6]

Question 2

Explain why i<=a.Length is unsafe in a loop indexing a[i], and repair. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Develop four distinct tests for a function counting even values, with expected answers and reasons. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A loop using i<=Length can safely read a[i] for every i.

Use key vocabulary from memory

1. Explain logic error in your own words.

2. Explain boundary in your own words.

3. Explain trace in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.